A new BioCommons infrastructure for the UK life science

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The ELIXIR-UK Node

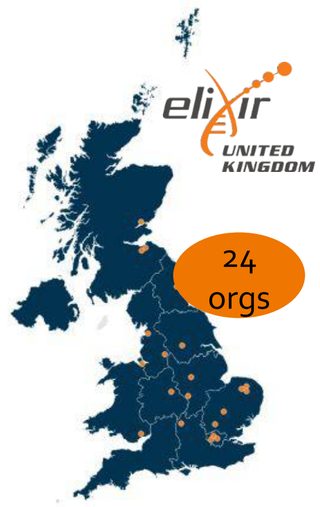
The national Node of ELIXIR in the United Kingdom, providing coordination and capacity building.

A consortium of 24 UK research organisations.

Supplies many of the ELIXIR Services to European and international partners.

Participates in a raft of ELIXIR Communities, activities and projects, leading many of them.

Works with UK funders on policy, infrastructure provision and skills.



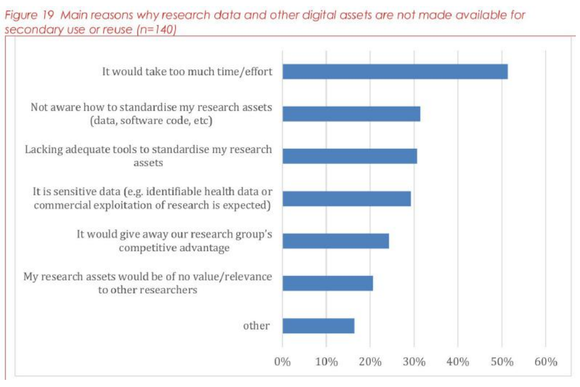
https://elixiruknode.org

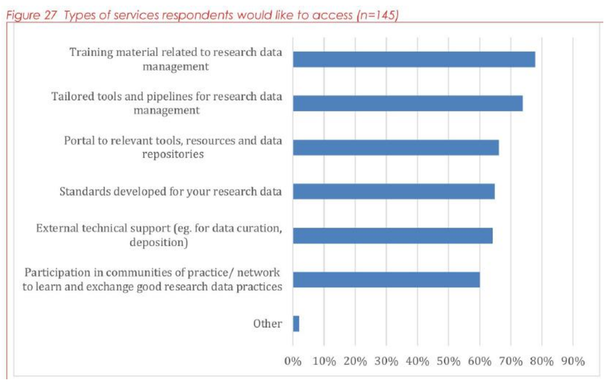
The need for a UK BioCommons

In 2021 ELIXIR-UK (together with Technopolis Ltd) undertook a 9-month **study** to:

- identify the **Research Data Management (RDM) needs** of UK bioscience researchers
- evaluate the **feasibility and impact** of the creation of a new UK Institute delivering a FAIR Data and Method BioCommons

This was funded by BBSRC, MRC and Wellcome, and included desk research, a community survey, dedicated interviews, and a technical analysis and scope.

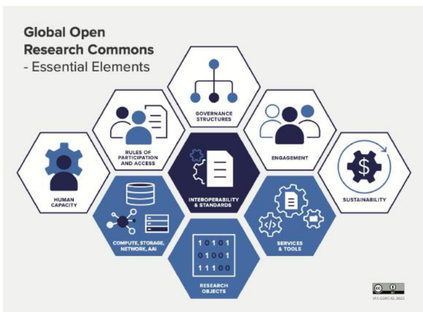




- Only **30%** of the UK life sciences community are compliant with the FAIR principles
- The study **recommended** the creation of a BioFAIR virtual institute to:
  - provide a **data commons** and a new **digital infrastructure** for the whole of the UK biosciences research community (20,000+ researchers)
  - support the adoption of FAIR RDM principles in order to produce a step-change in the sharing and re-use of research data and meaningful improvements in academic productivity, scientific quality and reproducibility and social benefits through greater knowledge spillovers.

International comparators

Australian BioCommons · NIH Common Fund Data Ecosystem · elixir Belgium · elixir Norway · elixir Germany · elixir Sweden

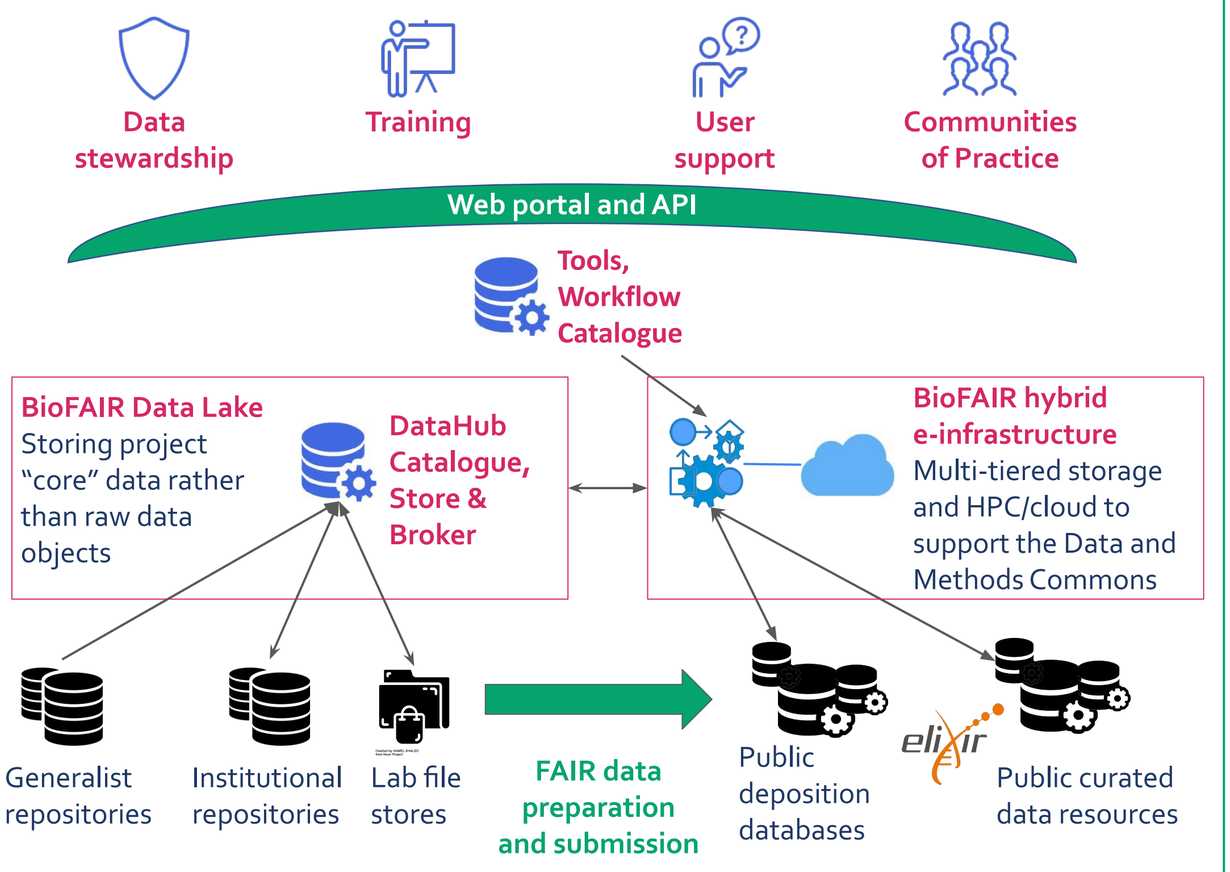


Global Open Research Commons (GORC) International Model developed by the RDA GORC IG.

Complementary research infrastructures

BBMRI.uk · UKRI NERC Environmental Data Service · Critical Mass UK · EuroBioImaging · HDRUK Health Data Research UK · PhenomUK · DSH · PSDI · DISSCO UK · EMBL-EBI

BioFAIR: a new Data and Methods Commons



BioFAIR Data Commons

- National **data catalogue** over a federated data lake
- Support **FAIR data at Research Performing Orgs** and project sites
- **Data brokering** to public archives
- Leverage ELIXIR's FAIR services, catalogues and data archives.
- FAIR Research Data Management advocacy

FAIRDOM · COPO · RDMkit · FAIRsharing.org · FAIRCOOKBOOK

BioFAIR Computational Methods Commons

- National **analysis capability** using computational workflows and notebooks
- Curate and share **best practice tools & workflows**
- Integration with data catalogue & stores
- **AI readiness** and QA
- Leverage ELIXIR's computational tools and workflow capability

WorkflowHub · jupyter · Galaxy · nextflow

BioFAIR Integrated Data Services

- **Metadata and Standards** framework based on **FAIR Digital Objects and APIs**
  - Bioschemas, RO-Crate, ISA; leveraging EU infrastructure developments.
- Life Science AAI Infrastructure (LS-Login)
- Data and method **stewardship** & User support
- **Communities of Practice**
- **Training** about the Commons and data stewardship; train-the-trainers
- One-stop shop web portal and APIs

Bioschemas · isatools · RO-Crate · LS LOGIN · elixir TeSS

Timeline

- 2018-2019: Discussions at ELIXIR-UK All Hands 2018, Management Committee retreat, ELIXIR-UK SIAB meeting and ideas finalised at UK All Hands 2019
- 2020: BBSRC, MRC and Wellcome awards ELIXIR-UK a grant for a feasibility study
- 2021: Feasibility Study published (with Technopolis) doi:10.5281/zenodo.7924339
- 2021: Initial UKRI Infrastructure Fund (IF) grant proposal
- 2022: UKRI IF announce an initial seed funding of £6.3 million over the next three years
- 2022: Submitted revised case for further funding to UKRI IF and Digital Research Infrastructure fund.

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